Computers in Biology and Medicine 184 (2025) 109342



Contents lists available at ScienceDirect

Computers in Biology and Medicine

journal homepage: www.elsevier.com/locate/compbiomed



Artificial intelligence-driven intelligent learning models for identification and prediction of cardioneurological disorders: A comprehensive study

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ARTICLE INFO

Keywords:
Artificial intelligence
Intelligent learning models
Cardiovascular
Cardio neurological disorders
Healthcare

ABSTRACT

The integration of Artificial Intelligence (AI) and Intelligent Learning Models (ILMs) in healthcare has transformed the field, offering precise diagnostics, remote monitoring, personalized treatment, and more. Cardioneurological disorders (CD), affecting the cardiovascular and neurological systems, present significant diagnostic and management challenges. Traditional testing methods often lack sensitivity and specificity, leading to delayed or inaccurate diagnoses. AI-driven ILMs trained on large datasets offer promise for accurate identification and prediction of CD by analyzing complex data patterns. However, there is a lack of comprehensive studies reviewing AI applications for the diagnosis of CD and inter related disorders. This paper comprehensively reviews existing integrated solutions involving AI and ILMs in CD, examining their clinical manifestations, epidemiology, diagnostic challenges, and therapeutic considerations. The study examines recent research on CD, reviews AI-driven models' landscape, evaluates existing models, addresses practical considerations, and outlines future research directions. Through this work, we aim to provide insights into the transformative potential of AI-driven ILMs in improving clinical practice and patient outcomes for CD.

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https://doi.org/10.1016/j.compbiomed.2024.109342
Received 11 May 2024; Received in revised form 19 October 2024; Accepted 30 October 2024
Available online 20 November 2024
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1. Introduction

The incorporation of Artificial Intelligence (AI) and Intelligent Learning Models (ILMs) in the healthcare domain has revolutionized the landscape by delivering unprecedented services such as precise diagnostics, remote health monitoring, personalized treatment, drug discovery, and chronic disease management [1,2]. The revolutionary influence of these technologies is facilitating the foundational support necessary for eradicating and managing diseases, ranging from minor ailments to life-threatening conditions, affecting trivial to vital body organs. The impact of these technologies in healthcare can be observed at three levels: for patients, it enables them to process their own data to promote health for larger benefits; for clinicians, it provides quick and precise image interpretation; and for healthcare systems, it enhances their capacity and potential by improving workflow and reducing human errors [3]. ILMs are advanced computational systems that utilize intelligent learning algorithms such as Machine Learning (ML) and Deep Learning (DL) algorithms to analyze vast amounts of data [4]. These models are designed to identify complex patterns and relationships within the data that may not be apparent through traditional analytical methods. By learning from a wide range of data inputs, including medical imaging, genetic information, physiological signals, and clinical records, ILMs can provide more accurate and personalized predictions and diagnoses. They continuously improve their performance by updating their algorithms with new data, making them increasingly effective over time in addressing various health conditions. Cardioneurological disorders (CD), encompassing a spectrum of medical conditions affecting both the cardiovascular and neurological systems (like brain stroke, syncope, heart failure, etc.), are the prominent reasons for hospital visits and admissions worldwide. Patients suffering with these disorders need longer hospital stays, bore expensive hospital charges, and are more susceptible to severe complications affecting overall prognosis and quality of life [5]. CD, due to their multifaceted nature, are typically associated with complex symptoms and variable disease progression, posing significant challenges for healthcare systems that are already grappling with the global shortage of staff [6]. Conditions such as cardiac arrhythmias, neurogenic orthostatic hypotension, and autonomic dysfunction represent a breadth of disorders within this diverse class, requiring a nuanced diagnostic and customized management strategy for the prevention and treatment of each such disorder. Traditional diagnostic methods often rely on individual clinical analysis and usual diagnostic tests, which may provide insufficient sensitivity and specificity, resulting in delayed or inaccurate diagnoses [7].

One specific study carried out to outline the frequency of diagnostic error for outpatient adults in the United States itself estimates that at least 5% of the results are erroneous, which is substantial to patients' safety risk [8]. AI-Driven ILMs trained over large datasets in this context pose themselves as the emerging potential solution for the near accurate identification and prediction of CD [9]. Using sophisticated computational algorithms, these models provide complex data patterns, including imaging studies, genetic markers, physiological signals, and clinical parameters, to differentiate disease signatures and their trajectories with better accuracy and precision [9]. Considering the socio-economic impacts and practical challenges involving the application of AI-Driven ILMs for the diagnostic and management of CD, various organizations and researchers are working tirelessly to come up with better and affordable solutions for such cases. As a result, there exists several services, prototypes, and applications in the domain. However, there is a lack of study which provides comprehensive review for CD involving AI or ILMs. Therefore, this work aims to explore the existing integrated solutions involving AI and ILMs in the broad domain of cardiovascular and neurological systems with specific focus on CD individually or combined. The study will facilitate the insight view of underlying principles, methodologies, and applications of these models to provide a comprehensive understanding of their revolutionary potential and transformative impact on clinical practice and patient outcomes.

The main objectives of this study include.

1. Examining the recent studies of CD, comprising its clinical manifestations, epidemiology, pathophysiology, diagnostic challenges, and therapeutic considerations.
2. Reviewing the landscape of AI-driven ILMs including ML and DL models along with their applicability in diagnosis, prognosis, and treatment of CD.
3. Evaluate and compare existing models for diagnosing, prognosing, and treating cardiovascular, and neurological, highlighting their strengths, challenges, and areas for improvement along with publicly available datasets and databases.
4. Addressing practical, ethical, and regulatory considerations comprising data privacy and model transparency in the deployment of AI-driven ILMs for clinical practice.
5. Highlighting the future direction and research focuses on clinical validation, real-time implementation, and multidisciplinary collaborations of AI-driven ILMs for the diagnosis, prognosis, and treatment of cardiovascular, neurological and CD.

The paper is organized as follows: Section 2 discusses the background and classification of CD. Section 3 explores the background of AI in healthcare. Various intelligent learning models for CD are covered in Section 4. Section 5 provides insights into commonly used databases and datasets for research. The statistical analysis of the impact of CD is presented in Section 6. Section 7 summarizes key findings and suggests future research directions. The paper concludes in Section 8.

2. Background of cardioneurological disorders

Cardioneurological Disorders constitute a group of conditions characterized by the coexistence of cardiac and neurological symptoms, often influencing one another. Fig. 1 provides a broader classification of CD, which divides the CD into two groups. One for cardiovascular and other for cardiac inflicted neurological disorders. Cardiovascular disorders (CVD) are further divided into coronary artery disease, arrhythmias, heart failure, and valvular heart disease. Coronary artery disease includes myocardial infarction (heart attack) and angina pectoris. Arrhythmias include Atrial Fibrillation (AF) and ventricular tachycardia. Heart failure is divided into systolic heart failure and diastolic heart failure. Valvular heart disease includes mitral valve prolapse and aortic stenosis. On the other hand, neurological disorders induced by cardiac dysfunctions include ischemic stroke, commonly caused by atherosclerosis and embolic stroke. Intracerebral hemorrhage is another disorder in this category, leading to hypertensive intracerebral hemorrhage and amyloid angiopathy. Vascular dementia and carotid artery disease also belong to this group; vascular dementia causes multi-infarct dementia and Binswanger's disease, while carotid artery disease leads to carotid artery stenosis and carotid artery dissection, respectively. Although epilepsy is primarily considered a neurological disorder, its inclusion in our paper is based on studies that suggest the onset of epileptic seizures can be predicted by ILMs using ECG signals that capture heart rate variability [10–12]. It may be noted that the figure provides a broader perspective on two groups of interrelated disorders and does not include an exhaustive list of all disorders within these categories.

These disorders highlight the complex interaction between the heart and the nervous system. Key genes like KCNQ1 and KCNH2 play pivotal roles, affecting both heart (ion channels) and brain function [13]. Such genetic mutations underscore the shared vulnerability, wherein patients may experience both cardiac issues (e.g., arrhythmias, prolonged QT) and neurological symptoms (e.g., seizures) [14,15]. Consequently, the overlapping symptoms pose diagnostic challenges, potentially leading to misidentification. To ensure accurate diagnosis and treatment, comprehensive assessments by both cardiologists and neurologists are

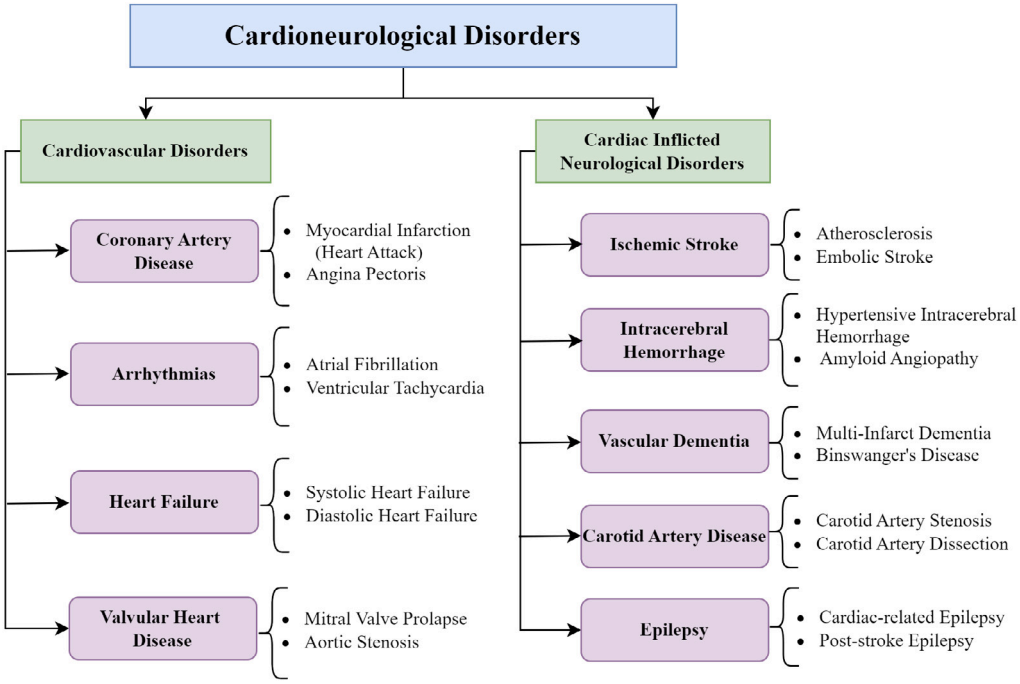


Fig. 1. Broad classification of cardioneurological disorders.

imperative. A holistic approach that considers both cardiac and neurological aspects is essential, as exemplified by the potential benefits of seizure control on cardiac health. Regular monitoring with EEG (Electroencephalography) and ECG (Electrocardiography) facilitates treatment adjustments as needed. Further research is necessary to fully understand the connection between the heart and neurological systems in CD. This deeper understanding could lead to more effective treatments and potentially preventative measures for such CD.

Table 1 provides a concise overview of the two types of disorders with respect to their development, how they typically present in patients, and the current medical interventions used to manage them.

Table 2 summarizes Long QT Syndrome (LQTS), a heart rhythm disorder with potential brain links. It explores two interpretations: neurological symptoms might be the root cause, or a consequence of heart issues. Both cardiac and neurological aspects are crucial for diagnosis and treatment due to overlapping symptoms and brain activity abnormalities in LQTS patients.

To understand the link between Epilepsy and Ictal Asystole, Table 3 explores three cases where epilepsy patients experienced a sudden stop in heartbeat (ictal asystole) during seizures [22]. All three patients benefited from pacemaker implantation, with some also requiring adjustments to their anti-seizure medication. The table highlights the potential link between epilepsy and heart rhythm disturbances.

On the other side, Table 4 discusses ictal asystole, a rare heart pause during seizures. It occurs in less than 0.5% of patients undergoing video EEG. This prevalence is slightly higher (0.32%) among those with hard-to-control epilepsy. While the most common seizure-related arrhythmia, it is still uncommon overall. The cause likely involves abnormal electrical brain activity disrupting heart control areas. There are usually no specific symptoms for ictal asystole itself, but the triggering seizure might cause typical seizure symptoms. Video EEG helps diagnose it. Treatment focuses on managing the epilepsy, with pacemakers a possibility in severe cases.

To understand the key imaging techniques for diagnosing heart and brain problems, Table 5 summarizes these two key imaging techniques by highlighting their advantages, disadvantages, and applications. CT scans offer a fast, comprehensive view with detailed pictures of bones, soft tissues, and blood vessels, but involve radiation exposure. Magnetic Resonance Imaging (MRI) provide detailed soft tissue images without radiation, making them ideal for brain and spinal cord issues, but they can be expensive and claustrophobic for some patients. Both play crucial roles in cardiology and neurology, with CT scans suited for blood vessel issues and MRIs excelling at brain and heart muscle assessments.

3. Background of artificial intelligence in healthcare

Early efforts in AI within the medical field, spanning from the 1960s to the 1980s, focused on rule-based expert systems. These systems were designed to assist with diagnosis and therapy decisions. Edward Shortliffe's MYCIN project at Stanford is a notable example from the 1970s; it utilized if-then rules in LISP programming to recommend antibiotics for infections [32]. Other significant systems from this period include INTERNIST, AI/RHEUM, and ONCOCIN. Despite their pioneering nature, these systems were challenging to create and adapt to new clinical situations. They operated by storing and organizing clinical knowledge in a knowledge base, which was then accessed through a user interface to provide advice [33–35].

Further developments in AI in medicine occurred at academic centres like Stanford, MIT, Rutgers, and Carnegie Mellon, with additional contributions from Europe. However, the integration of AI into clinical workflows was hindered by several factors including the slowness of tools and insufficient data. Over time, additional AI methodologies like fuzzy logic and neural networks (NN) began to emerge, enhancing the capability of AI to handle complex biomedical data and decision-making processes [36]. In recent years, there have been many research studies published combining AI with healthcare. To understand the evolution and applications, following subsections provide detailed analysis of different perspectives of AI in healthcare.

3.1. Applications of AI in healthcare

Artificial intelligence has been widely used in various applications within the healthcare sector, significantly enhancing medical practices and patient care. Table 6 showcases growing impacts of AI methods comprising ML and DL techniques alongside their precise

Table 1
Overview of cardiovascular disorders and neurological disorders.

Category	Pathogenesis (How they develop)	Clinical presentations (Symptoms)	Current treatments	Examples	Sources
Cardiovascular disorders	• Often involve chronic inflammation and damage to blood vessel walls. This can restrict blood flow and oxygen delivery to vital organs. • Risk factors include high blood pressure, high cholesterol, diabetes, smoking, and a sedentary lifestyle.	• Symptoms can vary depending on the affected organ system and severity. • Common signs include chest pain (angina), shortness of breath, fatigue, irregular heartbeats, and leg pain.	• Treatment plans typically involve a combination of lifestyle changes (diet, exercise, smoking cessation), medications (e.g., blood pressure medications, cholesterol-lowering drugs), and potentially minimally invasive procedures (angioplasty) or surgery (bypass surgery) to improve blood flow. • In severe cases, organ transplantation may be considered.	Coronary artery disease, heart failure, stroke (when caused by a blood clot blocking blood flow to the brain), peripheral arterial disease	[16–18]
Cardiac inflicted neurological disorders	• Chronic hypertension, atherosclerosis, blood clots from abnormal cardiac rhythms, and diminished blood flow all contribute to the development of cardiac-related neurological diseases. • These conditions reduce blood flow to the brain, which increases the risk of transient ischemic attack, strokes, and vascular dementia.	• Sudden weakness or numbness in the arms, legs, or face, especially on one side of the body, as well as disorientation or trouble speaking and understanding speech, are signs of neurological problems associated to the heart. • Severe headaches, dizziness, loss of balance or coordination, and vision issues in one or both eyes are also frequent. These signs frequently point to a transient ischemic attack or stroke. • Cognitive decline, memory loss, and concentration issues could be signs of vascular dementia.	• Cardiovascular-related neurological diseases are treated with anticoagulants to avoid strokes, antihypertensives to control blood pressure, and endarterectomy or carotid artery stenting to restore blood flow. • Modest lifestyle adjustments, such eating a heart-healthy diet and getting regular exercise, are essential for lowering risks and enhancing results.	Epilepsy, Atherosclerosis, Hypertension, AF.	[19,20]

Table 2
Understanding long QT syndrome: cardioneurologic vs. neurocardiologic perspectives.

Aspect	Description	Definition
Cardiac Manifestations	Primary Function Affected	LQTS disrupts the heart's electrical activity, causing prolonged QT intervals on ECG. This can lead to irregular heartbeats (palpitations), fainting (syncope), or even sudden death [13].
Genetic Predisposition	Underlying Cause	Mutations in genes like KCNQ1 and KCNH2, crucial for ion channel function in both the heart and brain, are often linked to LQTS. This genetic link might explain the co-occurrence of seizures and LQTS [14].
Cardioneurologic Viewpoint	Neurological Symptoms as Primary	This perspective suggests that neurological features (e.g., seizures) in LQTS patients might be the root cause, with cardiac symptoms developing secondarily. The presence of abnormal electrical activity in the brain (epileptiform activity) on EEG supports this view [15].
Neurocardiologic Viewpoint	Cardiac Symptoms as Primary	This viewpoint indicates that cardiac issues in LQTS are primary, with neurological symptoms as a consequence. For example, abnormal heart rhythms (arrhythmias) could lead to changes in blood flow to the brain (cerebral perfusion), affecting brain function [21].
Clinical Significance	Holistic Approach Needed	Given the potential overlap in symptoms, cardiologists and neurologists should consider both cardiac and neurological aspects when diagnosing and treating LQTS patients [21].
Research and Observations	Importance of Neurological Assessment	Studies show a high prevalence of EEG abnormalities in LQTS patients. This emphasizes the need for a broader diagnostic approach that may include neurological assessments [21].

application in diagnostics mechanisms of a diverse range of disorders. In contemporary research, these methods are being utilized to mitigate a multitude of challenges previously encountered with traditional approaches, thereby facilitating enhanced care delivery.

The criteria for selecting methods and AI applications presented in Table 6 are the type of data and the specificity of the disorder. The nature of the medical data significantly influences the choice of AI method, with CNNs excelling in image processing tasks like classifying skin cancer and brain tumors, while NLP is ideal for analyzing textual information. The specific medical condition also plays a crucial role, as different disorders require varying levels of accuracy and capabilities from AI models. For instance, SVMs are preferred for identification of heart valve diseases using heart sounds. due to their precision, and CNN-AlexNet are chosen automatic screening tool for cardiovascular diseases using different intervals of ECG signals. Focusing on these criteria ensures that AI applications are well-suited to the medical data and specific disorders, leading to more accurate and effective healthcare solutions.

Moreover, Table 7 summarizes research findings on the performance of AI models in diagnosing various CDs. Different methods, including CNN, random forest, SVM, long short-term memory (LSTM),

Table 3
Comparative summary of Ictal asystole cases in epilepsy patients.

Case	Age/Gender	Medical history	Symptoms	Diagnosis	Treatment	Outcome
1	40-year old male	Temporal lobe epilepsy (diagnosed at age 28)	Drop attacks, behavioral changes, memory lapses, focal seizures escalating to full-body convulsions	Ictal asystole (heart stoppage during seizures)	Dual chamber pacemaker	Improvement: No more drop attacks, but some behavioral issues and involuntary movements persist.
2	58-year old male	Newly diagnosed focal epilepsy	Brief lapses of consciousness, memory problems, difficulty speaking, sudden tiredness	Ictal asystole	Loop recorder to monitor ECG, permanent pacemaker, adjusted anti-seizure medication (levetiracetam replaced with sodium valproate and zonisamide)	Effective: No more lapses of consciousness, although some focal seizures remain.
3	60-year old male	No prior medical history	Brief lapses of consciousness, confusion, memory problems	Ictal asystole	Loop recorder to monitor ECG, dual chamber pacemaker, adjusted anti-seizure medication (levetiracetam replaced with lamotrigine)	Successful: No further seizures or fainting episodes after pacemaker placement, and lamotrigine controls remaining seizures.

Table 4
Key points on Ictal asystole: prevalence, causes, and management.

Parameter	Details	References
Epidemiology of Ictal Asystole	Ictal asystole is a rare event, occurring in less than 0.5% of patients undergoing video EEG for epilepsy diagnosis.	[23–27]
Prevalence in Specific Populations	The prevalence increases slightly (to 0.18%) among all patients referred for video EEG, and further rises to 0.32% in those with epilepsy that is difficult to control (refractory epilepsy).	[28,29]
Comparison to Other Cardiac Arrhythmias	While ictal asystole is the most common arrhythmia (irregular heartbeat) directly associated with seizures, it is still a rare occurrence overall in the epilepsy population.	[28]
Possible Causes	The exact cause of ictal asystole remains unclear, but it is believed to be linked to abnormal electrical activity in the brain regions controlling heart function.	[30]
Symptoms	Ictal asystole itself typically does not cause noticeable symptoms. However, the seizure that triggers it may present with usual seizure symptoms like loss of consciousness, jerking movements, or staring spells.	[30]
Diagnosis	Ictal asystole is often detected during video EEG, which simultaneously records brain activity and electrical activity of the heart (eg., ECG).	[24]
Treatment	Treatment for ictal asystole typically focuses on managing the underlying epilepsy with medications or surgery. In severe cases, a pacemaker may be implanted to regulate the heartbeat.	[31]

and autoencoders, were employed across different studies. The results varied across conditions, with high accuracy observed in detecting stroke (cerebrovascular accident), atherosclerosis, hypertension, epilepsy, brain hemorrhage, AF, vascular dementia, peripheral artery disease, and intracranial hemorrhage. Specific accuracies, sensitivity, specificity, and area under the curve (AUC) values are reported for each condition and method, highlighting the potential of AI-based diagnostic approaches in the field of neurology.

3.2. Role of AI in medical diagnosis and prognosis

Research suggests AI is shaking up medical diagnosis and prognosis. By analyzing vast amounts of medical data, AI algorithms can find hidden patterns that elude human experts. This translates to more accurate diagnoses, earlier disease detection, and better predictions of patient outcomes [65]. The paper likely explores the specific AI techniques used and the challenges of implementing them in healthcare, ultimately highlighting potential of AI to revolutionize medicine by improving diagnoses, optimizing treatments, and saving lives.

Table 8 shows how cardiologists and neurologists interact with medical AI tools. It explores their level of involvement in developing these tools, their comfort using them, and any anxieties they might have. The table also examines how these specialists perceive role of AI in their fields, considering aspects like improved patient outcomes, efficiency gains, and the importance of human oversight alongside capabilities of AI. By tailoring the measurement items to both cardiology and neurology, the table provides valuable insights into how medical AI is being integrated into these specialties.

3.3. Challenges of using AI in healthcare

Though AI in healthcare comes with significant potential but it is not free from challenges. On one hand, it promises to transform patient care through improved diagnosis accuracy, streamlined monitoring, faster drug development, and continuous virtual health support [71]. These advancements aim to create a more personalized and efficient healthcare system. However, ethical concerns regarding patient data

Table 5
Imaging the brain and heart: A look at key techniques.

Imaging method	Advantages	Disadvantages	Applications (cardio-neurology)
Computerized Tomography (CT)	• Painless, non-invasive, and accurate. • Images bones, soft tissues, enlarged ventricles, and blood vessels simultaneously.	• Exposes patients to radiation. • Not recommended for pregnant women.	• Imaging blood clots and abnormalities in blood vessels. • Assessing injuries or abnormalities in nerves or muscles of the eye.
MRI	• No radiation exposure. • Provides detailed images of soft-tissue structures, ideal for brain and spinal cord anomalies.	• Expensive. • May not be suitable for all patients due to claustrophobia or certain medical implants. • Cannot always distinguish between benign and malignant tumors.	• Detect subtle changes in brain and cardiovascular tissues, identify early signs of conditions like stroke, atherosclerosis, or vascular dementia. • Evaluating potential heart conditions, like enlarged ventricles or abnormal heart muscle function.
Echo	• Real time imaging • Provide comprehensive assessment of the heart including its size, shape, pumping capacity, and the function of its valves. • Obese patients or ones having lung disease may have difficulty obtaining clear echocardiogram images.	• Operator Dependent • Limitations in detecting certain abnormalities • Costly for those who need multiple tests.	• Provide size, shape, pumping capacity, and the function of heart valves. • Provide immediate assessment of heart's function and blood flow.
Nuclear Stress Test	• Highly sensitive in detecting coronary artery disease. • Early detection of abnormalities in blood flow to the heart muscle. • Gives objective and quantifiable information for diagnosis and risk stratification.	• Exposure to radiation. • Expensive • Limited availability as it requires specialized equipment and trained personnel.	• Early detection of abnormalities in blood flow to the heart muscle before the onset of symptoms. • Used to identify areas of reduced perfusion caused by blocked or narrowed coronary arteries.

Table 6
AI applications in CD.

Source	Method	Applications in Healthcare
[37]	Support Vector Machine (SVM)	Identification of heart valve diseases using heart sounds.
[38]	Convolutional Neural Network (CNN) - AlexNet	Automatic screening tool for cardiovascular diseases using different intervals of ECG signals.
[39]	ML, AI-powered systems	Diagnosis and prognosis of epilepsy, dementia, stroke, heart failure and arrhythmias.
[40]	AI-driven chatbots, ML, and DL models	Treatment and management of sudden cardiac arrest.
[41]	Capsule Network (CapsNet)	Coronary artery disease detection using ECG signals.
[42]	Deep Neural Network (DNN) with Discrete Wavelet Transform (DWT)	Automated detection of left bundle branch block from ECG signal.
[43]	SVM	Syncope identification and classification
[44]	SVM	Automated classification of hypertension and coronary artery disease patients using heart rate variability (HRV) analysis.
[45]	AI-transformer-based vision network model	Embolic stroke.
[46]	Natural Language Processing (NLP)	Identifying patients with cerebral aneurysms from text data.
[47]	AR and VR, AI-powered robotic, and AI-based emotional recognition systems	Rehabilitation and support (strokes or movement disorders, anxiety, allowing for early intervention).
[48]	CNN, RNNs, and DL models	Monitoring heart and brain health.
[49]	ANN, DL, and ML	Myocardial infarction (MI) and scar detection.

Table 7
Performance of AI in the diagnosis of CD.

Source	Condition	Method	Results
[50]	Stroke	Fused dual neural network	Accuracy of 98.19% in detecting Stroke.
[51]	Atherosclerosis	GoogLeNet	Accuracy of 98.20% with GoogLeNet.
[52]	Atherosclerosis	Neural network and Hierarchical clustering	Accuracy not specified in the study.
[53]	Epilepsy	Res-CNN	Achieved accuracy of 99% in epilepsy detection.
[54]	Epilepsy	LSTM	Achieved good binary classification of 95% and overall accuracy of 98%.
[55]	Vascular Dementia	CNN-VGG16, VGG19, DenseNet121, and InceptionResNetV2	Achieved an accuracy of 84.67% in Vascular Dementia diagnosis.
[56]	BrainHemorrhage	CNN, AlexNet, AlexNet-SVM	Achieved accuracy of 90% in hemorrhage detection.
[57]	Brain Hemorrhage	Autoencoders (Sparse auto-encoder, CNN)	Sparse auto-encoder achieved the best results in accuracy and error reduction.
[58]	Intracranial Hemorrhage	CNN (CNN-4 and VGG-16)	Achieved a range of AUC values from 0.50 to 0.98 for hemorrhage detection.
[59]	Atrial fibrillation	DL	Accuracy not specified in the study.
[60]	Atrialfibrillation	Light gradient boosting machine	Highest F1-score of 98.8%.
[61]	Hypertension	NLP-ALBERT	Highest F1 score of 95.0%.
[62]	Hypertension	Deep CNN	Achieved good performance with AUC 79.0% and better.
[63]	Peripheral artery disease	SMOTE and Enhanced Decision Tree (EDT) through HPO	Achieved accuracies of 99.20% and 98.52% respectively for identification of peripheral artery disease.
[64]	Peripheral artery disease	DL Enabled Algorithms	Achieved accuracy of 94.0% and AUC-ROC curve 97.0%.

privacy and biases in AI algorithms need attention. Additionally, ensuring AI technology reliability, increasing awareness and acceptance among healthcare professionals and the public, and managing professional liability risks are critical. To overcome these challenges, robust regulatory frameworks, reliable technological infrastructure, and comprehensive educational programs are necessary [72,73].

Table 9 provides an overview of the key challenges that need to be addressed to unlock the full potential of AI in transforming healthcare [74]. It serves as a strategic guideline in identifying critical challenges needed to be addressed before embracing the AI for its delivery of services. The provided summary includes multifaceted landscape of hurdles ranging from technical complexities to ethical considerations that demands attention and resolution for the optimal utilization of AI in healthcare domain.

4. Intelligent learning models for cardioneurological disorders

4.1. Traditional approaches vs. AI-driven models

While traditional methods handle smaller datasets and rely on established criteria, AI tackles vast amounts of complex data, improving diagnostic accuracy in intricate cases. The adaptability of AI allows for personalized treatments and faster integration of new research findings, allows a more individualized approach to cardiovascular and neurological care. Table 10 explores the major differences between traditional and AI-driven approaches in cardiology and neurology.

4.2. Use of machine learning techniques

Machine Learning is transforming the way we understand and manage CDs. This section explores various algorithms used for analyzing data like ECGs, brain scans, and genetics. Supervised learning methods like SVMs and random forests excel at classifying heart rhythms or identifying patterns in brain imaging data associated with specific diseases. Unsupervised learning methods like k-Means Clustering can group patients based on shared characteristics, while Reinforcement Learning holds promise for personalizing treatment strategies through real-time data analysis. Overall, these algorithms offer powerful tools for improving diagnosis, risk prediction, and ultimately, patient outcomes in cardioneurology. Fig. 2 presents a comprehensive list of ML techniques used in CD.

To better understand the above presented ML techniques, Table 11 explores various ML algorithms used in analyzing CDs by highlighting their learning type, examples and core function.

Table 12 presents an overview of ML techniques used in diagnosing various CVD. Different disorders, including myocardial infarction, cardiomyopathies, coronary artery disease, atherosclerosis, valvular heart disease, heart failure, and abnormal wall motion, are covered, along with the corresponding cohort sizes and ML algorithms employed. For myocardial infarction, SVM achieved high accuracy (0.99), while ANN and LR also demonstrated strong performance. Similarly, for cardiomyopathies, SVM and DT showed promising results. Pseudo-decision trees, ANN, and CNN were utilized for coronary artery disease, providing varying accuracies. CNN was employed for atherosclerosis with an

Table 8
Expanded measurement items for medical AI constructs.

Construct	Variables	Measurement items (cardiology focus)	Measurement items (neurology focus)	Additional parameters	Source
Medical Staff Participation	Level of Engagement	How involved am I in the development and feedback process for AI tools used in diagnosing and treating CVD?	How much input do I have in shaping AI tools for diagnosing and managing neurological disorders?	Training provided for effective participation	[66]
	Collaboration Quality	How effectively do I collaborate with developers on AI powered CVD diagnosis and treatment tools?	How well do I collaborate with developers on AI tools for neurological assessments and interventions?	Communication channels used (e.g., workshops, surveys)	
	Decision making Impact	To what extent can my voice influence key decisions regarding AI initiatives in cardiology?	How much say do I have in shaping AI applications for neurological care?	Level of decision-making authority	
AI Self Efficacy	Training Support	How well equipped am I to use AI tools for cardiovascular diagnoses after proper training?	How comfortable am I using AI technologies for neurological assessments following adequate training?	Availability and quality of training programs specific to cardiology/ neurology	[67,68]
	User Interface Accessibility	Can I easily access all necessary information through AI powered tools for CVD diagnosis and management?	Can I readily access the information I need through AI tools for neurological assessments and treatment planning?	Information clarity and organization specific to cardiology/ neurology practices	
	Perceived Ease of Use	How comfortable am I using AI technologies in my daily cardiology practice?	How straightforward is it for me to integrate AI tools into my workflow for neurological care?	Technical complexity of the AI system	
AI Anxiety	Specific Concerns	What aspects of AI use in cardiology make me feel uneasy? (e.g., reliance on algorithms, potential errors)	What specific features of AI in neurology cause me anxiety? (e.g., interpretability of results, job security)	Transparency and explainability of AI decision-making processes	[69]
	Job Security Impact	How concerned am I about AI replacing cardiologists in the future?	To what extent do I worry about AI replacing neurologists in the healthcare landscape?	Clear communication regarding the role of AI as a support tool	
	Perceived Control Loss	How worried am I about losing control over patient care decisions due to AI in cardiology?	To what extent does the use of AI in neurology make me feel like I'm losing control over patient care?	Level of human oversight and control maintained throughout the AI workflow	
Acceptance of Medical AI: Design Technologies	Patient Outcomes	How strongly do I support using AI to create personalized treatment plans for cardiovascular patients?	To what extent do I endorse AI assistance for accurate tumor identification before neurological surgery?	Evidence based data demonstrating improved patient outcomes with AI integration	[70]
	Efficiency Precision	How much do I value role of AI in improving the precision of presurgical planning for cardiovascular procedures?	How important is AI for streamlining surgical planning processes in neurology?	Efficiency gains and reduced error rates achieved through AI	
	Data Integration Innovation	How comfortable am I using big data analysis through AI to refine treatment protocols for CVDs?	To what extent do I advocate for role of AI in formulating treatment strategies for neurological disorders based on vast datasets?	Data privacy and security protocols in place for patient information	

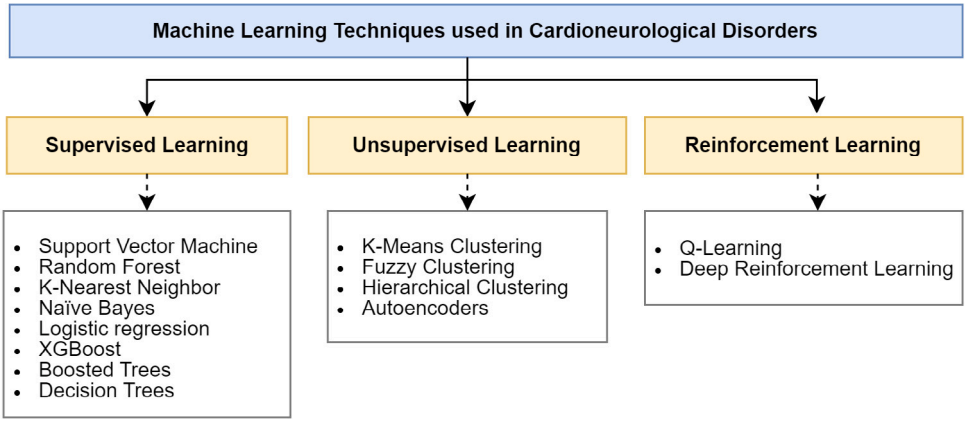


Fig. 2. Machine learning techniques used in CD.

Table 9
Key challenges in leveraging AI for healthcare transformation.

Challenge category	Description	Source	No. of studies	Percentage of studies addressing issue
Ethical and Privacy Concerns	• AI in healthcare raises questions about patient data privacy, potential biases in algorithms, and the ethical considerations of using AI for diagnosis or treatment. • Building trust and ensuring transparency are critical.	[75–78]	4	26.70%
Awareness and understanding	• A significant gap exists between the public and healthcare professionals regarding capabilities of AI and limitations. • This knowledge gap can hinder trust in AI solutions and their adoption in practice.	[71,79]	2	13.30%
Reliability and Trust	• Concerns about the reliability and trustworthiness of AI systems are a major hurdle. • Both patients and healthcare providers need to be confident in the accuracy and effectiveness of AI for it to be fully integrated into medical care.	[71,80–83]	5	33.30%
Professional Liability and Integration	• The integration of AI into healthcare practices raises questions about professional liability. • How will responsibility be shared between healthcare providers and AI systems for medical decisions and patient outcomes?	[81,84–86]	4	26.70%

Table 10
Traditional vs. AI-driven approaches in cardiology and neurology.

Parameter	Cardiology	Neurology
Data Handling	**Traditional:** Analyzes smaller datasets, relies on manual data interpretation (e.g., blood tests, ECG printouts). **AI:** Manages large, complex datasets, integrating diverse data types (ECG recordings, imaging results, patient demographics).	**Traditional:** Handles structured data (clinical observations, MRI scans) based on established criteria. **AI:** Analyzes vast amounts of brain imaging data (MRI, CT scans) and genetic information to identify patterns.
Accuracy	**Traditional:** Achieves high accuracy for well-understood conditions (e.g., heart failure) based on established criteria. **AI:** Improves diagnostic accuracy in complex cases (arrhythmia detection) by analyzing ECG data with high precision.	**Traditional:** Relies on clinician expertise, leading to some variability in accuracy due to subjectivity. **AI:** Enhances prediction of disease progression based on ECG, brain scans and genetic data, allowing for earlier intervention.
Adaptability	**Traditional:** Static, relies on slow updates from new research findings. **AI:** Continuously learns and adapts based on new data, making it adaptable to personalized treatment approaches.	**Traditional:** Slow to integrate new research findings into clinical practice. **AI:** Adapts readily to emerging treatments and diagnostic tools, paving the way for personalized medicine.
Examples	**Traditional:** Uses ECGs, stress tests, and blood tests to diagnose heart disease. **AI:** Utilizes ML models to analyze ECG data for predicting heart failure with greater accuracy, enabling earlier intervention.	**Traditional:** Relies on clinical assessments, MRI scans, and established criteria for diagnosing multiple sclerosis. **AI:** Employs DL models to detect subtle changes in brain structure from MRI scans, aiding in the early diagnosis of CD.

Table 11
ML algorithms used in cardioneurological disorders.

Algorithm	Learning type	Description	Examples in cardioneurology	Considerations
SVM	Supervised	SVM excels at classifying different heart rhythm patterns (arrhythmias) on ECG data. It can identify subtle differences between normal and abnormal cardiac activity.	• Classifying ECG recordings to detect arrhythmias like AF or ventricular tachycardia. •Differentiating between various types of seizures based on EEG data.	• Requires careful selection and pre-processing of ECG/EEG data for optimal training. • Tuning hyperparameters is crucial for accurate classification.
K-Nearest Neighbor (KNN)	Supervised	KNN can be used to predict the risk of heart failure based on patient characteristics. It classifies new patients based on the similarity to existing data points with known outcomes.	• Predicting the likelihood of heart failure progression based on medical history and vital signs. • Stratifying patients with neurological disorders based on disease severity using clinical data.	• Accuracy can be affected by the size and quality of the training data. • Choosing the optimal number of neighbors (K) is important.
Random Forest	Supervised	Random Forest is a powerful tool for analyzing brain imaging data. It combines multiple decision trees to identify patterns in MRI scans associated with neurological disorders.	• Classifying MRI scans to detect Stroke. • Identifying potential biomarkers for neurological conditions based on brain imaging features.	• Requires a substantial amount of labeled MRI data for training. • Tuning hyperparameters can be complex.
Naı̀ıve Bayes (NB)	Supervised	Naı́ıve Bayes can be used to analyze genetic data and identify mutations linked to an increased risk of CD.	• Identifying genetic variants associated with a predisposition to heart disease or stroke. • Analyzing genetic data to assess a patient's risk of developing a specific CD.	• Assumes independence between genetic features, which may not always be true. • Performance can be impacted by the quality and completeness of genetic data.
CNN	Supervised	CNNs are adept at analyzing medical images like coronary angiograms to detect blockages in blood vessels. They excel at identifying patterns in complex images.	• Analyzing coronary angiograms to identify and segment areas of stenosis (narrowing) in arteries. • Detecting subtle abnormalities in brain structure from ECG and MRI scans for early diagnosis of CD.	• Requires large datasets of labeled medical images for effective training. • Computational cost of training CNNs can be significant.
Autoencoder (AE)	Unsupervised	Autoencoders can be used to identify hidden patterns in large datasets of medical records, potentially leading to new insights into cardioneurological conditions.	• Identifying subgroups of patients with similar disease characteristics for targeted treatment approaches. • Extracting relevant features from complex medical data to improve diagnostic models.	• Reconstruction errors in the autoencoder can influence downstream tasks. • Unsupervised nature requires careful selection of data representations.
K-Means Clustering	Unsupervised	K-Means Clustering can be used to group patients with CD based on shared characteristics.	• Identifying clusters of patients with similar stroke risk factors for targeted preventive measures. • Grouping patients with neurological conditions based on clinical presentation or treatment response.	• Choosing the optimal number of clusters (K) is crucial for meaningful results. • Sensitive to outliers and the initial selection of cluster centroids.
Reinforcement Learning	Dynamic Programming	Reinforcement learning has the potential to optimize treatment strategies for CD by iteratively improving decision-making based on patient response.	• Optimizing medication regimens for heart failure patients based on real-time physiological data.	• Requires careful design of reward functions and exploration/exploitation strategies. • Computational cost and interpretability of results can be challenging.

accuracy range of 0.77 to 0.8. SVM exhibited high accuracy (0.99) for valvular heart disease, while SVM also showed good performance for heart failure and abnormal wall motion. Overall, this data suggests that ML holds significant potential for improving heart disease diagnosis.

4.3. Use of deep learning techniques

Under the umbrella of ILMs, DL is revolutionizing cardiovascular disease diagnosis using ECG data [106]. Researchers are exploring various techniques, from general diagnosis to specific conditions like arrhythmias. CNNs, LSTMs, and even combinations are used to analyze ECGs from wearables, single leads, and even time-frequency image representations. While accuracy and efficiency are high for some models, challenges like noise sensitivity and interpretability remain. Table 13 presents a snapshot of this promising field, highlighting its potential and the need for further development to ensure reliable clinical applications.

4.4. Use of explainable AI techniques

In literature, various ILMs have been developed to diagnose the CVD efficiently. In particular, ML with various explainable AI (XAI) techniques provide potential solutions. Table 14 shows how researchers are making DL models used for CVD diagnosis with ECG data more transparent. It explores the different ML algorithms, XAI techniques used, and the insights gained from these explanations.

4.5. Use cases of ILMs in disorders

Though AI is being applied to a wide range of healthcare scenarios, particularly in diagnosing, predicting, and managing cardiovascular and neurological disorders. In certain cases, specific AI methods are more frequently employed due to their effectiveness and suitability for the type of data and nature of the disorder. Table 15 illustrates the common applications of AI techniques in these disorders, highlighting the specific AI methods most frequently used in each case.

Table 12
Machine learning techniques in diagnosing cardiovascular disorders.

Cardiovascular Disorders	Cohort size	ML Algorithms (Accuracy/AUC))	References
Myocardial Infarction	160, 800, 180	ANN - 0.94 SVM - 0.99 CNN - 0.89 RNN - 0.84 XGBoost - 0.97 LR (AUC) - 0.92 DeepMI (AUC) - 0.94	[87],[88–91]
Cardiomy-opathies	83	DT - 0.79 CNN - 0.78 LR - 0.67 Random forest 0.72 SVM - 0.92, SVM (linear) 0.69, SVM (RBF) 0.67, Additional SVM - 0.86 Boosted trees 0.75 Neural-Networks 0.74	[92–96]
Coronary Artery Disease	267, 115, 58, 308	Pseudo-DT - 0.8 Multiple Pseudo-DT variants (0.77 & 0.73 accuracy) ANN - 0.74	[97–99]
Atherosclerosis		CNN - Accuracy range 0.53 - 0.77 RCNN - Accuracy range 0.63 - 0.8	[100]
Valvular Heart Disease	102	SVM (RBF) - 0.99 LDA+NN - 0.96	[101]
Heart Failure	397	SVM - AUROC range 0.70–0.76 DQN – 0.99	[102,103]
Abnormal Wall Motion	20, 58	SVM (Linear) - Accuracy range 0.87 - 0.95 SVM (RBF) - Accuracy range 0.81 - 0.85	[104,105]

5. Databases and datasets for research

In order to harness the power of AI and to revolutionize its diagnostics capabilities, various databases and datasets are discussed in this section. It highlights a description of the database, dataset, the AI models applied over it and the reported performance of the models in terms of their accuracy measures.

5.1. Database

Table 16 provides an overview of several biological databases relevant to cardioneurological research, along with their descriptions, links, and applications in research due to space constraints, the links are given below the table at the referred link number. These databases include the Gene Expression Database, TCGA (The Cancer Genome Atlas), KEGG (Kyoto Encyclopedia of Genes and Genomes), BioCyc, ZINC, and PubChem. The Gene Expression Database offers gene expression data from various tissues, aiding in uncovering genes linked to disease development and progression in conditions like heart disease and stroke [144]. TCGA provides genomic data, indirectly relevant to cardioneurology, assisting in identifying genetic variants potentially linked to increased risk of CD [145]. KEGG offers intricate maps of biochemical pathways, facilitating insights into the functional roles of genes implicated in cardioneurological conditions. BioCyc provides curated pathway data, aiding in visualizing interactions between genes and proteins and identifying potential bottlenecks for therapeutic intervention. ZINC houses a vast collection of chemical compounds, useful for virtual screening to identify potential drug candidates for CD. However, limitations on purchasable compounds and bioactivity information may necessitate further exploration [146,147]. Lastly, PubChem offers comprehensive information on chemical structures and properties, aiding in gaining a deeper understanding of existing cardioneurological drugs and their mechanisms [148]. These databases serve as valuable resources for researchers in the field of cardioneurology, offering diverse applications in their research endeavors.

5.2. Datasets

Table 17 summarizes the key features of some datasets used in ML research on heart disease. By consolidating the pool of information, the table aims to facilitate insights into the datasets used by various researchers and organizations for the diagnosis of CD.

Table 18 presents various patterns observed in heart disease datasets along with the ML algorithms explored and the reported accuracies in different studies.

In [149], a dataset comprising 303 patients with cardiovascular disease and 14 features was explored using RIPPER, DT, ANN, and SVM achieving an accuracy of 81.08%. McManus et al. in [150] utilized a cohort study of 1246 individuals with 28 features, applying NB algorithm and achieving an accuracy of 83.19%. A dataset of 303 individual Rajaie Cardiovascular cases with 34 features was analyzed using Bagging, C4.5 classification algorithms, yielding accuracies of 79.54% and 68.96%, respectively [151]. Enterprise Data Warehouse & Research Patient Data Repository datasets were used in [152], where Logistic Regression (LR), Gradient Boosting (GB), Max-out networks, and DUNs were employed, achieving accuracies ranging from 73.9% to 76.4%. Zhang et al. in [153] combined data from multiple sources comprising 920 cases and 76 features, exploring Bagged DT, Gradient Boosting DT, KNN, RF, SVM, LR, achieving accuracies of up to 96.6%.

The Cleveland heart dataset with 14 features in [154] was analyzed using C4.5, RF, MP, BN, and NB, with accuracies ranging from 79.87% to 84.16%. In [155], the Careggi University Hospital dataset comprising 2303 instances and 38 features was explored using RF, SVM with linear

Table 13
Overview of deep learning models for cardiovascular disease analysis.

Sources	Primary CVD focus	Learning model	Architecture description	Key metrics (accuracy/ AUC)	Data source	Model advantages	Model limitations
[107]	Arrhythmia Diagnosis	Deep CNN	3 convolutional layers binary classification	0.8568 (ACC)	1-lead ECG	Improved accuracy with lead data manipulation	Limited arrhythmia coverage; dataset-dependent results
[108]	ECG Quality Assessment	CNN-LSTM with CGAN	2 CNN branches, 1 LSTM branch; data augmentation with CGAN	0.971 (ACC), 0.986 (SENS), 0.964 (SPEC)	1-lead ECG	Effective generalization diverse data generation	Limited to quality assessment; does not quantify signal quality
[109]	ECG Anomaly Detection	CNN with feature selection	Feature selection via recursive feature elimination	0.889 (SENS), 0.902 (F1-score)	12-lead ECG	Efficient diagnosis with feature selection	Performance affected by ECG noise
[110]	Long QT Syndrome (LQTS) Detection	CNN with XceptionTime modules	Spatial–temporal analysis with XceptionTime	0.911 (Balanced ACC)	12-lead ECG	Robust validation across diverse groups	Needs broader clinical validation; potential control group bias
[111]	Heart Rate Recovery (HRR) Analysis	DenseNet-based 1D CNN	Predicts impaired HRR	0.483 (Pearson correlation)	1-lead ECG	Predicts HRR for clinical outcomes; uses resting ECG	Modest correlation; potentially better with more leads
[111]	Arrhythmia Diagnosis	Parallel cross-convolutional neural network (RNN-CNN)	Combines RNN and 2D CNN	0.997 (ACC)	1-lead ECG (Lead II)	Efficient for imbalanced data; excellent learning of ECG features	High training time due to large parameter set
[112]	Arrhythmia Diagnosis	1D CNN with Attention Branch Network	Multi-loss function optimization	0.846 (Multilabel ACC), 0.812 (F1-score)	12-lead ECG	Addresses multilabel classification and explainability	Potential efficiency reduction due to zero-padding
[113]	Arrhythmia Diagnosis	BiLSTM with tree regularization	Feature relevance optimization with tree regularization	0.993 (ACC)	1-lead ECG (Lead II)	Superior regularization for explainability	Limited details on interpretability provided
[114]	AF Diagnosis	2D convolutional BiLSTM	Uses time-frequency ECG images	0.991 (ACC, SENS, SPEC)	2-lead ECG	Fast and efficient AF detection; uses spatial–temporal correlations	
[115]	Blood Pressure Estimation	Regression	1D CNN with dilated strided convolutions	5.32 (MAE systolic), 3.38 (diastolic)	Raw ECG (PPG) Signals	End-to-end model; adaptable to various signals	Limited by low sampling rate and ICU data
[41]	Coronary Artery Disease	Capsule Network	ECG-specific adaptations	0.994 (2-sec), 0.986 (5-sec)	Short ECG segments	Effective for small datasets; robust to attacks	Higher computational demand than CNNs
[38]	General CVD Diagnosis	Residual Network (ResNet)	Variant with shortcut connections	0.998 (ACC), 0.995 (SENS), 0.999 (SPEC)	12-lead ECG	Minimal preprocessing; handles imbalanced data	Limited disease specificity due to short signal analysis
[116]	CVD Diagnosis via ECG Analysis	Convolutional Bi-LSTM with attention time-adaptive CNN	Average ACC: 0.989	2-lead ECG signal	Utilizes time-adaptive input; highly accurate	Overfitting; high memory demand	
[117]	Arrhythmia Diagnosis	Multi-channel CNN with depth-wise separable convolutions	Average ACC: 0.994	3-lead ECG	Efficient parameter usage; superior for arrhythmias	Requires signal preprocessing; high computational load	
[118]	Arrhythmia Detection from ECG Images	Inception-V3 based DeepECG system	Multiple convolution layers	0.984 (Bal ACC), 0.954 (SENS), 0.967 (SPEC)	1-lead ECG (Lead II) images	Handles 2D ECG images; suitable for mobile devices	Performance varies with image quality; lacks noise filtering
[119]	AF Detection	Neural network with dilated causal convolutional blocks	0.986 (ACC), 0.987 (SENS), 0.990 (SPEC)	1-lead ECG signal	Fast and accurate; suitable for real-time applications	Performance affected by data imbalance	

(*continued on next page*)

Table 13 (*continued*).

Sources	Primary CVD focus	Learning model	Architecture description	Key metrics (accuracy/ AUC)	Data source	Model advantages	Model limitations
[120]	AF Detection (Wearable Devices)	1D CNN with HRV features; transfer learning	0.955 (ACC), 0.953 (AUC)	ECG and PPG signals	Adapts ECG models for wearables; uses transfer learning	Overlapping class samples; ignores non-linear HRV features	
[87]	Myocardial Infarction Analysis	LSTM-based network with spectral longitudinal analysis	0.967 (normal MI) to 0.738 (old MI)	12-lead ECG	Comprehen-sive spectral-temporal analysis; large dataset handling	Ineffective in distinguishing among leads	
[121]	Cardio Monitoring in COVID-19 Patients	CNN and LSTM combination	0.992 (ACC), 0.977 (SENS), 0.995 (SPEC)	ECG from wearables	Low latency, high throughput; robust model generaliza-tion	Limited heartbeat type data; potential SMOTE bias	
[122]	R Peak Detection in ECG	U-Net with Inception and Residual blocks	0.9837 (F1-score)	1-lead ECG signal	Effective R-peak detection with minimal post-processing	Reduced effectiveness at low SNR levels; high computational cost	
[123]	Arrhythmia Diagnosis	1D CNN with stacked residual blocks	0.966 (Average ACC), 0.971 (AUC)	2-lead ECG	Model ex-plainability; identifies top-performing leads	Possible population bias; needs further validation	

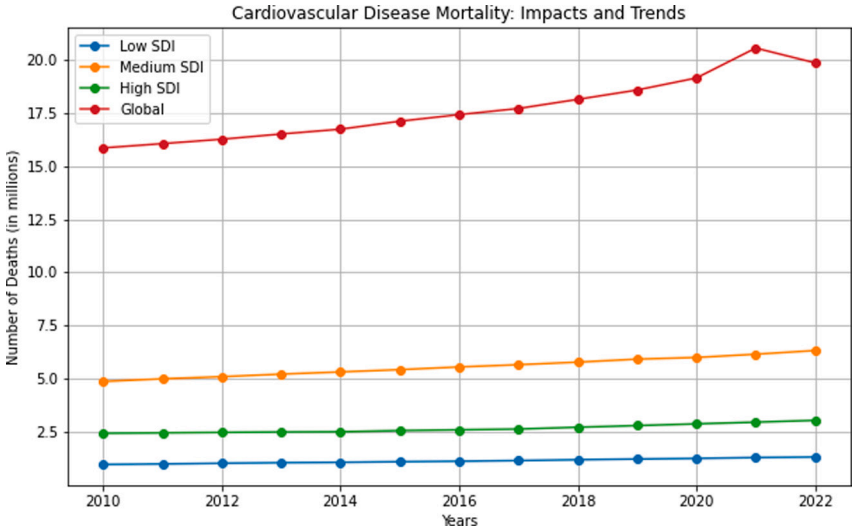


Fig. 3. Age standardized mortality impacts and trends for CD across global SDI strata.

and radial basis function (RBF) kernels, Boosted Trees, and NN, achieving accuracies ranging from 67% to 75%. Lastly, Ye at al. in [140] utilized a dataset of 1767 heart failure with preserved ejection fraction (HFpEF) patients from 4 countries, applying LR with forward selection & lasso regularization, RF, Gradient Boosting Decision Trees (GDB), and SVM with linear and radial basis kernels, achieving accuracies ranging from 70.3% to 76%.

6. Statistical analysis

Considering the severity of the disease it is important to have clear insights about the trends and figures involving the CD. Figs. 3 to 11 highlights various trends and measures for CD from years 2010 to 2022 [86].

Fig. 3 depicts age standardized impacts on mortality across global Socio-Demographic Index (SDI) strata. It can be observed that the number of deaths due to CD increased from 15.85 million in 2010 to 19.84 million in 2022. Fig. 4 shows the age standardized disability-adjusted life year (DALYs) trends across global SDI strata. It can be inferred from the figure that the global burden of CD is continuously increasing and it is a major contributor to the global burden of disease. Fig. 5 represents the age standardized gender specific mortality trends, which represent almost equal distribution into the number of deaths for both male and females. Fig. 6 highlights age standardized global burden of mortality and DALYs involving strokes. A careful view of the figure points out that the number of deaths and DALYs are both increasing continuously during the years and thus putting a significant burden on the healthcare industry. Subsequently, age standardized gender specific

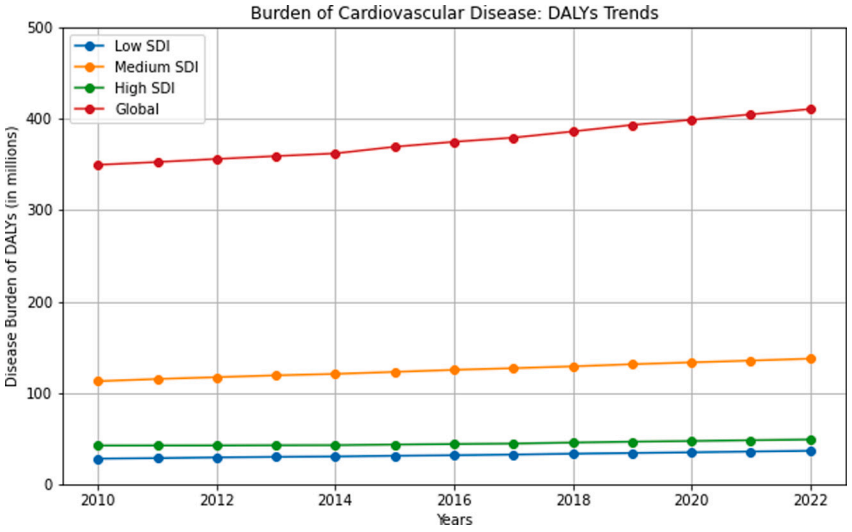


Fig. 4. Age standardized DALYs trends for CD across global SDI strata.

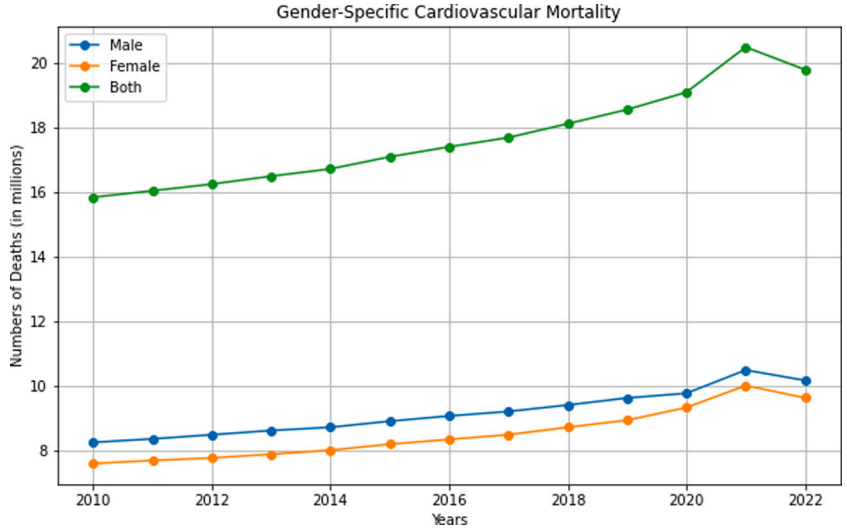


Fig. 5. Age standardized gender-specific mortality trends for CD.

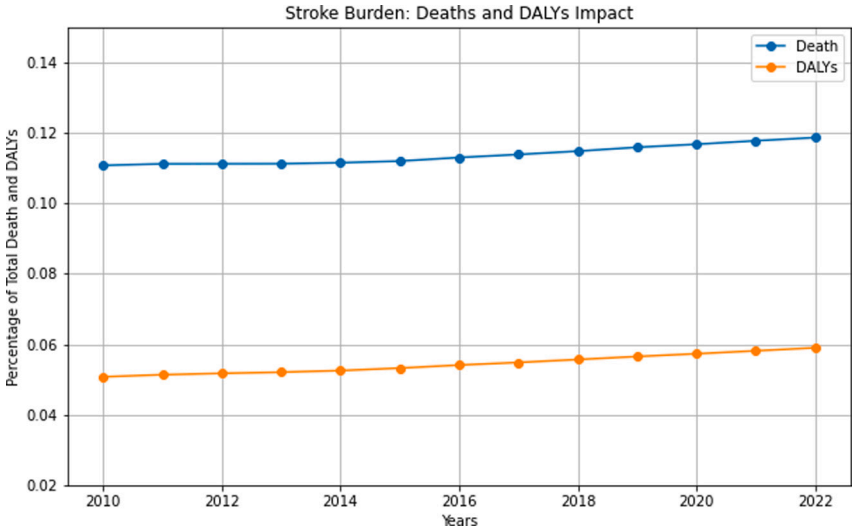


Fig. 6. Age standardized global burden highlighting mortality and DALYs trends for strokes.

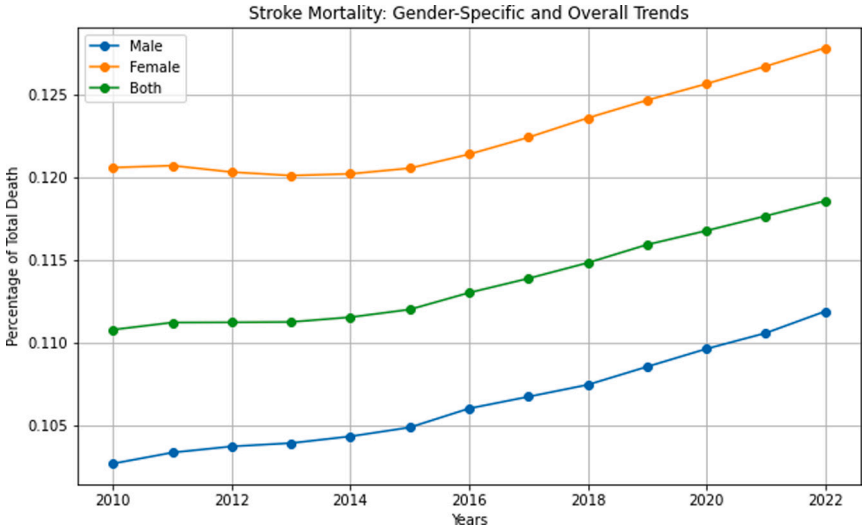


Fig. 7. Age standardized gender specific mortality trends for strokes.

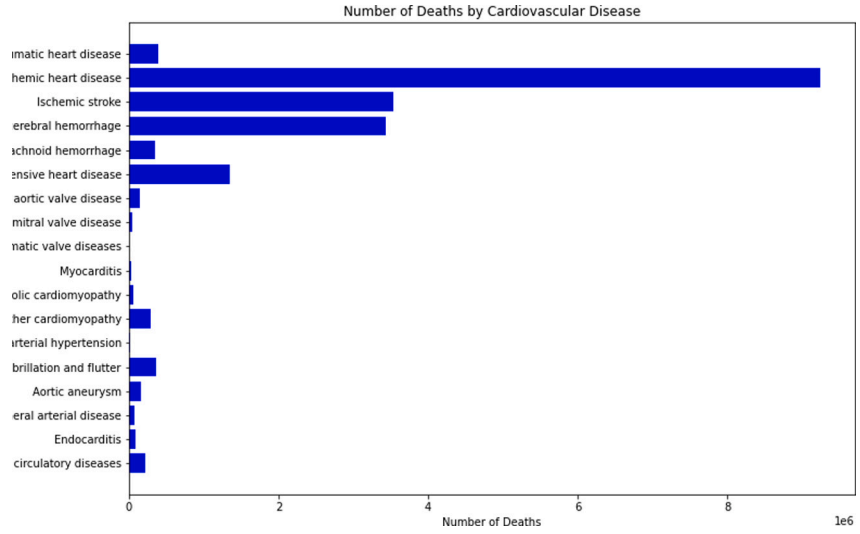


Fig. 8. Age standardized mortality counts involving all cardiovascular disorders.

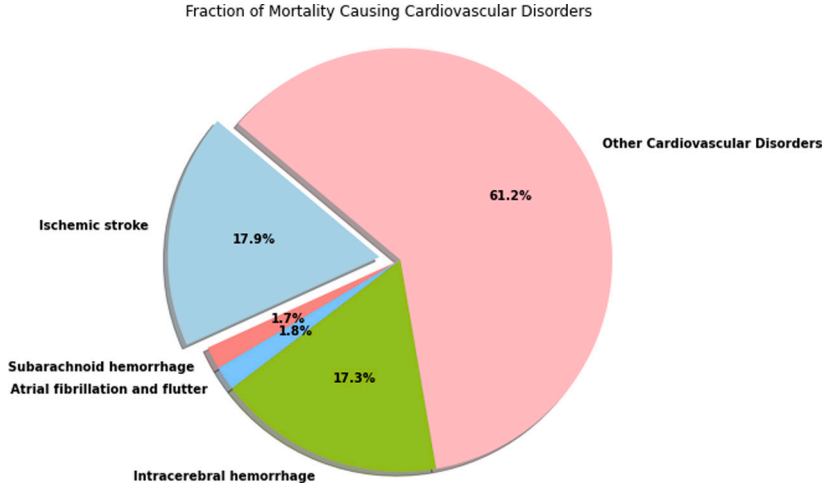


Fig. 9. Percentage of mortality involving CD Vs. Overall cardiovascular disorders.

Table 14
Overview of intelligent learning methods with explainable AI techniques.

Sources	ILM algorithm	XAI technique	Explanation scope	Key findings
[124]	CNN	Class Activation Mapping and Attention	Local (within a patient's ECG)	Highlights specific ECG segments for age prediction, revealing each lead's contribution and pinpointing segments suggesting a younger or older rhythm.
[106]	CNN	Gradient CAM	Local (within a patient's ECG)	Identifies the most influential parts of the ECG affecting the diagnosis, helping pinpoint the location of abnormalities.
[125]	CNN	Gradient CAM	Local (within a patient's ECG)	Emphasizes key ECG areas crucial for diagnosing a specific type of fast heart rhythm (wide QRS complex tachycardia).
[111]	CNN	Saliency Maps	Global (across patients)	Outlines the influential ECG areas for heart rate recovery predictions, summarizing data from a large group of individuals.
[126]	CNN	SHAP	Local and Global	Provides a two-level explanation: highlighting important ECG regions for each patient and assessing the overall importance of different ECG leads across all classifications.
[123]	CNN	SHAP	Local and Global	Offers detailed explanations at both individual and population levels, including insights into misclassifications.
[127]	RF	Feature Importance	Global (across patients)	Evaluates the most important ECG features and demographic data, identifying the factors with the strongest influence on the model's predictions.
[128]	RF	SHAP	Global (across patients)	Explains feature importance and reveals interactions between features (e.g., how age and sex interact with QTc duration and hypertension) in various classification scenarios.
[129]	SVM	SHAP	Global (across patients)	Quantifies the impact of each feature on ECG signal classification, also highlighting potential issues with the dataset.
[130]	SVM	Surrogate Model (Decision Tree)	Global (across patients)	Uses a simplified decision tree model to analyze the importance of ECG features in AF diagnosis.
[88]	XGBoost	SHAP	Local and Global	Showcases influential features for diagnosing acute myocardial infarction, including explanations for specific patient cases.
[131]	XGBoost	SHAP	Global (across patients)	Identifies the top features impacting AF prediction, based on average SHAP values.
[112]	CNN	Class Activation Mapping	Local (within a patient's ECG)	Visualizes critical ECG segments during multi-label classification (classifying multiple conditions at once), with review from cardiologists for validation.
[113]	CNN	Decision Tree	Global (across patients)	Develops a simulated decision tree to clarify the model's decision-making process based on critical ECG points.
[132]	CNN	Human-in-the-Loop	Agnostic (no specific technique)	Combines human expertise with AI insights for a more flexible and interpretable analysis of ECG signals. This allows for adjustments based on real-world knowledge.
[133]	CNN	Class Activation Mapping	Local (within a patient's ECG)	Focuses on ECG segments indicating abnormal patterns linked to arrhythmias.
[134]	SVM	Feature Importance	Global (across patients)	Identifies key features for coronary artery disease classification using a method called Weight-by-SVM.

mortality trends for strokes is given in Fig. 7, which shows that rate of death causing strokes are more prevalent in females than male. In order to present a combined number of age standardized death counts involving all CVD, Fig. 8 shows a bar chart depicting number deaths counts related to disorders categorized in cardiovascular disease. Consequently, Fig. 9 shows a pie chart depicting a fraction of mortality causing CD to total mortality due to CVD. It can be observed from the chart that the neurological disorders yield a significant fraction of deaths in comparison to the total death counts. Fig. 10 again presents a bar chart illustrating age standardized prevalence counts for disorders that can be categorized into CVD. Results show that the number of people affected with this disorder is rising continuously and over 520 million people are affected worldwide. This surge is putting a major strain on healthcare expenditure, and also adversely impacting the quality of life for millions. Fig. 11 in the last revisits the prevalence of CD vs overall CVD. It highlights that a quarter of the population affected due to CVD are also suffering from CD.

7. Opportunities and future directions for AI in healthcare

As the field of CD continues to evolve, it is crucial to identify future directions that hold promise for advancing research, diagnosis, and treatment. This section outlines potential avenues for exploration and development in CD research, drawing insights from domain-specific opportunities and challenges, diverse opportunities for AI in healthcare, and research area-wise potential research directions.

Table 19 provides an overview of domain-specific opportunities and challenges within the area of CD. By examining the unique characteristics and complexities of CD, researchers can identify key areas for further investigation and innovation.

Table 15
Typical use cases of ILMs in used in disorders.

Disorder category	Disorder type	Typical use cases	ILM method	Source
Cardiology	Arrhythmia	Detection and classification of arrhythmias from ECG data	DL (CNNs, RNNs), ML (SVM), Bi-LSTM, IQPSO-SVM, Bi-LSTM, Fusion NN	[107,113,113,117, 130,132,133]
	Heart Failure	Predicting risk and early diagnosis	DL (ANNs), ML (Logistic Regression, Decision Trees), NLP	[135–137] [92,138,139]
	Coronary Artery Disease	Diagnosis using coronary angiograms, risk prediction	DL (CNNs), ML (KNN, Random Forest), ID-CADCapsNet, NN, Fuzzy Clustering	[41,98,99,134]
	Myocardial Infarction	Automated detection and prediction from imaging and ECG data	DL (CNNs, RNNs), ML (SVM), DeepMI	[87,88]
	Hypertension	Monitoring and risk prediction	DL (ANNs), ML (Logistic Regression, Random Forest)	[140]
Neurology	Ischemic Stroke	Identification of blockages in blood vessels, guiding treatment plans, and assessing brain damage	ML, DL	[51,141]
	Intracerebral Hemorrhage	Typically used to describe cases of bleeding within the brain caused by conditions such as hypertensive intracerebral hemorrhage and amyloid angiopathy.	DL (CNN), ML (SVM, KNN)	[142]
	Vascular Dementia	Managing patients with multi-infarct dementia and Binswanger's disease	DL (RNNs, ML SVM)	[143]
	Carotid Artery	Diagnosing and managing conditions like carotid artery stenosis, carotid artery dissection, and assessing stroke risk.	DL (CNNs), ML (SVM)	[141]

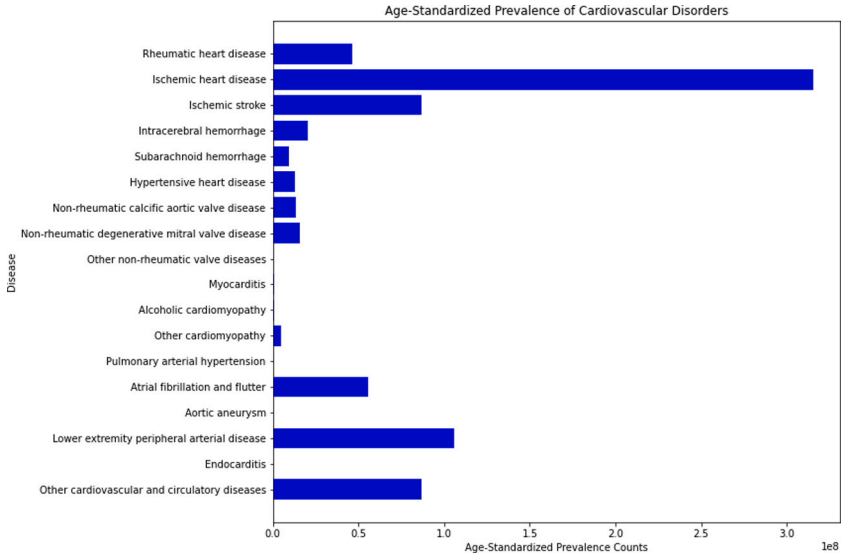


Fig. 10. Age standardized prevalence counts involving all cardiovascular disorders.

Table 16
Biological databases for cardioneurological research.

Database	Description	Link	Cardio- neuro relevance	Applications in research
Gene Expression Database	Contains RNA-Seq, microarray data from tissues, including brain and heart. Requires familiarity with gene IDs or keywords.	L1	Yes	• Identify genes linked to diseases like heart disease, stroke, neurodegeneration. • Analyze expression changes in response to treatments.
TCGA (Cancer Genome Atlas)	Primarily cancer-focused, but includes genomic data useful for identifying mutations related to cardioneurology.	L2	Indirectly	• Find genetic variants linked to conditions like Alzheimer's or inherited heart diseases. • Requires expertise in navigating cancer-centric data.
KEGG (Kyoto Encyclopedia of Genes and Genomes)	Maps of biochemical pathways for genes and proteins. Provides functional insights and therapeutic targets.	L3	Yes	• Discover functional roles of genes in cardioneurology. • Use pathway analysis for therapeutic target discovery.
BioCyc	Curated pathway data focused on metabolic pathways relevant to humans.	L4	Yes	• Visualize gene-protein interactions in pathways like calcium signaling in the heart. • Identify targets for therapeutic intervention.
ZINC	Collection of purchasable chemical compounds with limited bioactivity information.	L5	Limited	• Screen for potential drug candidates. • Explore chemical similarities to existing drugs.
PubChem	Database of chemical structures and properties. Useful for understanding drug mechanisms.	L6	Yes	• Gain insights into cardioneurological drug mechanisms.

[L1] https://www.ncbi.nlm.nih.gov/geo/
[L2] https://www.cancer.gov/ccg/research/genome-sequencing/tcga
[L3] https://pubmed.ncbi.nlm.nih.gov/10592173/
[L4] https://www.sri.com/platform/biocyc-database-collection/
[L5] https://zinc15.docking.org/
[L6] https://pubchem.ncbi.nlm.nih.gov/

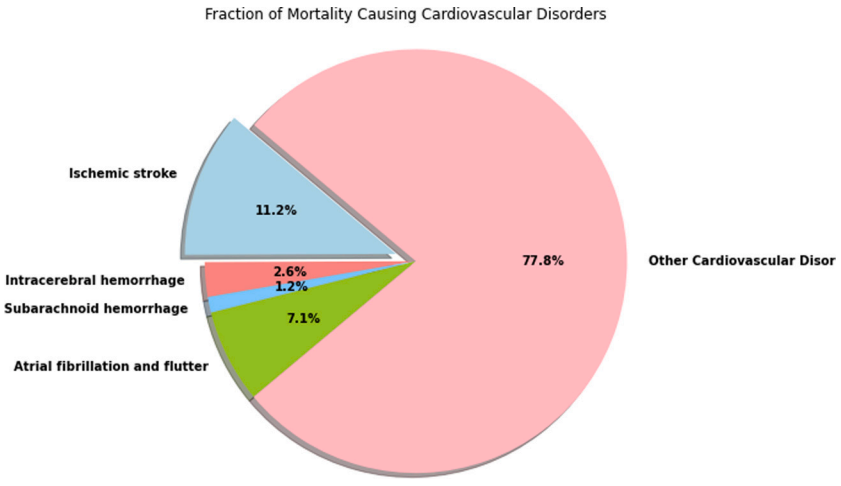


Fig. 11. Percentage of prevalence involving CD Vs. Overall cardiovascular disorders.

Table 20 presents diverse opportunities and future directions for the application of AI in healthcare, specifically focusing on its potential impact on the diagnosis and management of CD. It covers improvements in clinical teamwork, advancements in medical technology, enhancements in diagnosis and monitoring, and the acceleration of drug development. Additionally, it highlights role of AI in providing virtual health assistance, predictive health analytics, and surgical precision.

Following are the potential research directions within specific areas of CD research. By exploring these research areas and identifying gaps in current knowledge, researchers can chart a course for future investigations that may lead to novel insights and breakthrough discoveries in the field of CD.

Vascular Calcification

- Explore underlying causes: Investigate genetic, metabolic, and inflammatory pathways that contribute to vascular calcification [169–171].
- Specificity of location: Study the role of calcification in different arterial beds (intracranial, coronary) and its link to cardiovascular risks [172,173].

Table 17
Public dataset for CD research.

Dataset name	Description	Link	Focus
Stroke Prediction Dataset	This dataset predicts stroke risk based on input parameters like gender, age, diseases, and smoking status, with each row containing relevant patient information.	L7	Stroke
OpenNeuro	Open-access platform for sharing neuroimaging data.	L8	Various CDs
Prediction of atherosclerosis using machine learning based on operations research	The study analyzed 622 patient samples, identifying 304 as atherosclerosis risk group and 318 as controls, with 25 features retained after data analysis.	L9	Atherosclerosis
Hypertension Prediction	This dataset contains 13 feature variables, 1 target variable, and has balanced classes.	L10	Hypertension
Bonn University EEG Dataset	Collection of EEG (electroencephalogram) data for neurological research.	L11	Various Neurological Disorders (EEG)
Bern-Barcelona Dataset	Another collection of EEG data for neurological research.	L12	Various Neurological Disorders (EEG)
AF Termination Challenge Database	This two-channel ECG dataset was created for the 2004 Computers in Cardiology Challenge, aimed at predicting spontaneous AF termination.	L13	AF
Vascular Dementia Prediction Dataset	This dataset includes 373 T1-weighted MRI scans from 150 right-handed subjects, aged 60 to 96, with each subject scanned across two or more visits over at least a year.	L14	Vascular dementia
Open Access Series of Imaging Studies	Cross-sectional MRI data in young, middle aged, nondemented and demented older adults.	L15	Various CDs
Cleveland Clinic Heart Disease Dataset	This dataset contains 303 observations with 13 features from non-invasive diagnostic tests and patient information, plus 1 target attribute.	L16	Peripheral Artery Disease
RSNA Intracranial Hemorrhage Detection	Four research institutions provided de-identified CT studies for the challenge dataset, labeled by ASNR volunteers, with support from MD.ai, and RSNA aims to use AI to enhance hemorrhage detection and expedite clinical work.	L17	Intracranial-Hemorrhage
BigBrain	High-resolution 3D MRI dataset of a human brain.	L18	Brain Anatomy Research
Australian Imaging, Biomarker Lifestyle Flagship Study of Ageing	Imaging and biomarker data related to aging research.	L19	Aging and Neurological Disorders
Epileptic Seizure Recognition	The dataset consists of 500 EEG recordings, each with 4097 data points over 23.5 s, which were divided into 23 chunks of 178 data points each, resulting in 11,500 rows of 1-second recordings labeled with 5 classes.	L20	Epilepsy
Ischemic Stroke Lesion Segmentation	Dataset for training algorithms to segment ischemic stroke lesions in brain imaging.	L21	Ischemic Stroke

[L7] https://test.ieee-dataport.org/documents/stroke-prediction-dataset
[L8] https://openneuro.org/
[L9] https://www.aimspress.com/data/article/preview/pdf/62302d02ba35de7cf5703fae.pdf
[L10] https://www.kaggle.com/datasets/prosperchuks/health-dataset/data?select=hypertension_data.csv
[L11] https://www.ukbonn.de/
[L12] https://www.upf.edu/
[L13] https://www.physionet.org/content/aftdb/1.0.0/
[L14] https://www.kaggle.com/datasets/shashwatwork/dementia-prediction-dataset?select=dementia_dataset.csv
[L15] https://www.oasis-brains.org/
[L16] https://figshare.com/articles/dataset/processed_cleveland_data_csv/20410665?file=36493260
[L17] https://www.kaggle.com/c/rsna-intracranial-hemorrhage-detection/data
[L18] https://bigbrainproject.org/
[L19] https://aibl.org.au/
[L20] https://www.kaggle.com/datasets/harunshimanto/epileptic-seizure-recognition?select=Epileptic+Seizure+Recognition.csv
[L21] http://www.isles-challenge.org/

Table 18
Accuracies on machine learning algorithms on heart disease datasets.

Study	Dataset description	ML algorithms explored	Reported accuracy
[149]	303 patients with cardiovascular disease (14 features)	RIPPER, DT, ANN, SVM	81.08%
[150]	Cohort study of 1,246 individuals (28 features)	NB	83.19%
[151]	303 individual Rajaie Cardiovascular cases (34 features)	Bagging, C4.5 classification	79.54%, 68.96%
[152]	Enterprise Data Warehouse Research Patient Data Repository	LR, GB, Max-out networks, DUNs	73.9%–76.4% (varies by source algorithm)
[153]	Combined data from multiple sources (920 cases, 76 features)	Bagged DT, Gradient Boosting DT, KNN, RF, SVM, LOR	Up to 96.6% (varies by algorithm)
[154]	Cleveland heart dataset (14 features)	C4.5, RF, Model Pruning, BN, NB	79.87%–84.16% (varies by algorithm)
[155]	Careggi University Hospital dataset (2303 instances, 38 features)	RF, SVM - linear RBF, Boosted Trees, ANN	67%–75% (varies by algorithm)
[140]	Dataset of 1,767 heart failure with preserved ejection fraction (HFpEF) patients from 4 countries	LR with forward selection lasso regularization, RF, Gradient Boosting Decision Trees, SVM - linear radial basis kernels	70.3%–76% (varies by algorithm)

Table 19
Domain wise opportunities and challenges.

Domain	Successful applications	Opportunities	Challenges
Signal processing	• Automated detection of AF using smartwatch data [149,150]. • Identifying ventricular fibrillation and tachycardia to improve shock delivery in automated defibrillators (AEDs) [151]. • Classifying heart abnormalities based on ECG and other signals like seismocardiography (SCG) and gyrocardiography (GCG).	• Growth of wearable devices and smartwatches. • Large public datasets available for research. • High accuracy achieved in some applications.	• Gap between development and real-world use in hospitals. • Ethical considerations around privacy, data ownership, and cost-effectiveness. • Potential to widen healthcare disparities.
Image processing	• Automated analysis of echocardiograms, including image interpretation, segmentation, and disease detection [152,153]. • Volumetric and functional assessment of heart MRI scans [154]. • ML models for fractional flow reserve (FFR) as an alternative to complex simulations [155].	• Large public datasets available for training algorithms. • Easy implementation by medical device vendors. • Potential to reduce human error in image interpretation.	• Lack of transparency in some vendor algorithms. • Significant variability among doctors interpreting images. • Models may not generalize well to different patient populations.
Clinical risk stratification	• Predicting the development of high blood pressure [140]. • PREDICT software for cardiovascular risk assessment [156]. • GO-FAR score for heart failure risk [157]. • HF2PEF score for heart failure with preserved ejection fraction [158].	• Integration with electronic medical records (EMRs) for point-of-care decision making. • Early identification of patients at high risk.	• Limited access to complete patient data in some healthcare systems. • Need for real-time updates to models as data evolves. • Balancing accuracy with interpretability of risk scores for doctors.
Natural language processing	• Congestive heart failure treatment information extraction system (CHIEF) [136]. • Algorithm for identifying heart failure cases in a health information exchange network [137].	• Automating data extraction from electronic health records (EHRs). • Voice-based information capture to improve patient-doctor communication.	• Balancing accuracy with generalizability across different healthcare settings and doctor documentation styles. • Manual verification and annotation of data can be time-consuming.
Fundamental clinical discoveries	• Identifying patient subgroups with distinct characteristics and treatment responses using phenomapping analysis [159,160]. • Integrating omics data (genomics, proteomics, etc.) into treatment planning and drug development [77].	• Propelling research advancements and improving cardiovascular disease management. • Deeper understanding of disease mechanisms and patient variability.	• Difficulty in interpreting complex models with multiple layers of data. • Challenges in data annotation, storage, and computational resources.

Table 20
Diverse opportunities and future directions for AI in healthcare.

Opportunity in healthcare	Impact of AI	Number of studies	Studies	Future research directions	Enhancements for cardio & neuro disorders
Enhanced Teamwork and Decision Making	Improved collaboration and more accurate clinical decisions	2	[72,73]	Explore AI integration within multi- disciplinary teams	• Develop AI tools that facilitate communication and information sharing between specialists. • Design AI systems that can learn from the combined expertise of a team.
Pioneering Technological Advances	Development of innovative tools for treatment and care	5	[135,161–164]	Develop AI-powered diagnostic devices	• Leverage AI for non-invasive imaging analysis in cardiology (e.g., echocardiograms) and neurology (e.g., EEGs). • Create AI-powered tools for personalized treatment plans in cardiology (e.g., heart failure management) and neurology (e.g., seizure prediction).
Advanced Diagnosis and Patient Monitoring	Increased precision in diagnosis and real-time patient monitoring	4	[165–168]	Implement AI in remote patient monitoring systems	• Develop AI algorithms to analyze medical wearables data for early detection of cardiac issues (e.g., arrhythmias) and neurological disorders (e.g., tremors). • Design AI-powered systems for continuous monitoring of neurological function post-stroke or brain injury.

Intracranial Internal Carotid Artery

- Different calcification types: Distinguish the contributions of intimal (inner lining) and medial (middle layer) calcification to stroke risk and outcomes [173].
- Stroke details: Examine the relationship between intracranial artery calcification and stroke subtypes, recurrence, and mortality, considering diverse populations [172,174].

Coronary Artery Calcification

- Risk prediction beyond atherosclerotic cardiovascular disease (ASCVD): Refine the use of coronary artery calcification scores to predict overall ASCVD risk, including coronary heart disease, stroke, and mortality [171].
- AF and coronary artery calcification: Investigate the link between coronary calcification and AF, considering age-related differences and disease progression [175].

Clinical Implications

- Risk assessment tools: Translate research findings into practical tools for doctors. Develop risk assessment models that incorporate arterial calcification as a risk factor for stroke, AF, and other cardiovascular complications [176].
- Treatment strategies: Evaluate the effectiveness of interventions targeting vascular calcification to reduce cardiovascular morbidity (illness) and mortality (death) [177].

Cardioneurological disorders

- Investigate the potential link between arterial calcification and the development or progression of CD, such as cognitive decline, dementia, and migraine.
- Explore if arterial calcification can be used as a biomarker to identify individuals at higher risk of these conditions.
- Examine whether treatments targeting vascular calcification can also improve neurological outcomes.

Incorporating insights from above presented information, researchers and healthcare practitioners can chart a course for future research endeavors aimed at addressing the multifaceted challenges posed by CD and advancing the development of innovative diagnostic and therapeutic strategies.

8. Conclusion

In this paper, we explored CD and explored the imperative need for precise diagnostic mechanisms to effectively manage these complex medical conditions. With the rising incidence of CD worldwide, there is an urgent need for precise diagnostics mechanisms to improve patient outcomes and reduce healthcare burdens. Through an in-depth examination of the literature, case studies, and epidemiological data, we highlighted the critical importance of accurate diagnosis and prognosis in improving patient outcomes and quality of life. The evolution of AI has driven a new era in healthcare, offering innovative solutions to longstanding challenges in disease detection and management. We discussed the evolution and diverse applications of AI in healthcare, emphasizing its pivotal role in enhancing medical diagnosis and prognosis. Despite the immense potential of AI, we also highlighted the inherent challenges and opportunities associated with its implementation in clinical practice. Furthermore, we explained the significance of ILMs in the context of CD, contrasting traditional approaches with AI-driven methodologies. Through a comprehensive review of ML techniques employed in CD identification, we highlighted the transformative impact of AI on diagnostic accuracy and efficiency. Moreover, we explored the role of datasets and databases in facilitating AI-driven diagnostic models for CD, emphasizing the importance of high-quality, diverse datasets in training and validation. By leveraging large-scale datasets, AI algorithms can discern intricate patterns and relationships within patient data, enabling more accurate and personalized diagnostic insights. In conclusion, this paper underlines the actual importance of AI-driven ILMs in transforming the diagnosis and management of CD. By incorporating the power of AI, healthcare practitioners can unlock new possibilities for early detection, personalized treatment, and improved patient outcomes in the domain of CD. Future

research in AI-driven healthcare should validate predictive models and ILM-driven diagnostic algorithms through prospective, multicenter studies with diverse patient groups. Integrating genetic information and environmental exposures can enhance predictive accuracy. Assessing AI effectiveness in clinical practice, addressing ethical challenges, and ensuring proper training of ILMs are crucial for improving cardiovascular disease care and shaping future diagnostics and treatment.

CRediT authorship contribution statement

Shahadat Hussain: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Shahnawaz Ahmad:** Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Mohammed Wasid:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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